Faculty of Engineering
The University of Hong Kong

CIVL1118 STRUCTURAL MECHANICS

Course Outline

Course code:	CIVL1118
Course title:	Structural Mechanics
Offered by (department):	Department of Civil Engineering
Level:	1
Credit units:	6
Pre-requisite course code:	ENGG1000 Engineering Fundamentals
Co-requisite course code:	Nil
Learning hours:	Lecture (35 hours) Tutorial (4 hours) Laboratory (9 hours)
Student quota:	Each class limited to 100
Course type:	Core
Offer in academic year:	1st and 2nd semesters
Assessment (%):	Written examination (60%) Continuous assessment (30%) Practical work (10%)

This document contains information for the course CIVL1118 Structural Mechanics.

Its intention is for students undertaking the course to be well informed in terms of its learning expectations and also to help make their learning journey an enjoyable one.

Information presented in this document was correct at the time of printing.

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Course Description and Aims

The primary objective is to provide students with an understanding of the behaviour of beams and simple structures, which is required for their future studies in structural and geotechnical engineering. Students will develop the skills of evaluating the performance of engineering structures of various types under external loads within the elastic limit and analysing the vulnerability of these structures to various actions. This course covers stress and strain; generalized Hooke's law; static determinate beams and frames; torsion; combined loadings; flexural deformation of beams; stress transformation and Mohr's circle; failure criterion; column buckling. In addition to a number of lectures on the course topics, there will be laboratory work on buckling of columns. The buckling test aims to show Euler's buckling load for slender columns. Students in small groups will conduct the experiments together but they will analyse the test data and submit the report independently.

At the end of the course, students who fulfil the requirements of this course will be able to:
1. Understand fundamental concept of stress, strain and their relationship
2. Understand thermal effects on axial loading
3. Analyse static determinate beams and frames subjected to external loads
4. Analyse normal stress of unsymmetric section and composite section
5. Analyse shear stress due to shear force and evaluate shear centre location
6. Understand torsion of circular sections and thin-walled tubes
7. Understand stress and strain transformation for 2D problems
8. Understand deformation of beams due to bending moment
9. Understand principles of column buckling
10. Analyse failure of structures subjected to various criteria

Teaching Staff

Contact details for the course co-ordinator and course teachers are shown below:

Course Co-ordinator	
Name (Department: Specialty)	**Contact Details**
Prof. J.J. Wang (Civil Engineering: Structural Engineering)	E-mail: cewang@hku.hk Phone: 3917 2647 Office: Room 6-04 Haking Wong Building
Course Teachers	
Name (Department: Specialty)	**Contact Details**
Prof. H. Ye (Civil Engineering: Structural Engineering)	E-mail: hlye@hku.hk Phone: 3917 8132 Office: Room 6-20 Haking Wong Building
Prof. J.J. Wang (Civil Engineering: Structural Engineering)	E-mail: cewang@hku.hk Phone: 3917 2647 Office: Room 6-04 Haking Wong Building

Learning Outcomes and Alignment with HKIE Attributes for Engineering Graduates

By the end of this course students should be able to demonstrate a threshold level of mastery of the following learning outcomes. The appropriate HKIE abilities associated with each of the learning outcomes are also presented.

Course Learning Outcomes	**Relevant HKIE Attributes for Engineering Graduates (Equivalent to Programme Learning Outcomes, POs)**
1. Analyse stress-strain relationship	PO(b) identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development; PO(e) create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to support detailed analysis of complex engineering problems, and recognise limitations of such techniques, resources and tools;
2. Understand thermal effects on axial loading	PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex engineering problems;

Course Learning Outcomes	Relevant HKIE Attributes for Engineering Graduates (Equivalent to Programme Learning Outcomes, POs)
3. Analyse internal force of statically determinate beams and frames under loads and draw internal force diagrams	PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex engineering problems.
4. Analyse normal stress of unsymmetric section and composite section	PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex engineering problems.
5. Analyse shear stress in beams and understand shear centre	PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex engineering problems.
6. Analyse torsion of circular sections and thin-walled tubes	PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex engineering problems; PO(e) create, select and apply appropriate techniques, resources, and modern engineering and IT tools, including prediction and modelling, to support detailed analysis of complex engineering problems, and recognise limitations of such techniques, resources and tools.
7. Understand section properties and polar moment of area	PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex engineering problems.
8. Understand 2D stress and strain transformation	PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex engineering problems.
9. Understand deflection of beams	PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex engineering problems.
10. Understand principles of column buckling	PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex engineering problems; PO(b) identify, formulate, research literature and analyse complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences and engineering sciences with holistic considerations for sustainable development.
11. Analyse failure of structures subjected to various criteria	PO(a) apply knowledge of mathematics, natural science, computer science and engineering appropriate to the degree discipline to develop solutions to complex engineering problems.

Course Teaching and Learning Activities and Alignment with Learning Outcomes

Teaching and learning activities in this course include lectures, tutorials, assignments and laboratory works. All course learning outcomes will be introduced and explained through lectures, tutorials and laboratory works.

The course contents and its alignment with the course learning outcomes and assessment tasks are described below.

Course Content	Responsible Lecturer	Course Learning Outcomes
Concept of stress, strain and their relationship	Prof. H. Ye	1
Thermal effects on axial loading	Prof. H. Ye	2
Internal force and moment in static determinate beams and frames	Prof. H. Ye	3
Section properties and parallel axes theorem	Prof. H. Ye	7
Beams in symmetric bending	Prof. H. Ye	4, 7
Normal stress of unsymmetric section and composite section	Prof. J.J. Wang	4,7
Shear stress due to shear force and shear center	Prof. J.J. Wang	5
Torsion of circular sections and thin-walled tubes	Prof. J.J. Wang	6
Stress and strain transformation for 2D problems	Prof. J.J. Wang	8
Flexural deformation of beams	Prof. J.J. Wang	9
Column buckling	Prof. J.J. Wang	10

Course Content	Responsible Lecturer	Course Learning Outcomes
Failure criterions	Prof. J.J. Wang	11
Weeks 1-13 Laboratory Work		
Test on buckling of columns	Prof. J.J. Wang	10

Guidelines on Grading and Standards of AssessmentD

HKU has adopted standards-based assessment. The general course grade descriptors aim to provide descriptions on the normal expectations of student achieving particular grades.

Course Grade	Description
A	Exceptionally good performance demonstrating a superior understanding of the subject matter, a foundation of extensive knowledge, a skillful use of concepts, and ability to analyze and evaluate problems.
B	Good performance demonstrating a good understanding of the subject matter, more than adequate knowledge, capacity to use the appropriate concepts, and an ability to handle moderately difficult problems.
C	Adequate performance demonstrating an adequate understanding of the subject matter, adequate knowledge, an ability to handle relatively simple problems, and adequate preparation for moving on to more advanced work in the field.
D	Minimally acceptable performance demonstrating at least partial familiarity with the subject matter and some capacity to deal with relatively simple problems, but also demonstrating deficiencies serious enough to make it inadvisable to proceed further in the field without additional work.
F	Unacceptable performance demonstrating unfamiliarity with the subject matter, and lack of capacity to deal with relatively simple problems, and also demonstrating deficiencies serious enough to make it advisable to retake the course.

Course Assessment Tasks and Alignment with Learning Outcomes

Assessment tasks in this course are described below, which includes weighting, assessment type, and alignment with course learning outcomes.

Weighting of continuous assessment and written examination

Assessment Type	Percentage of Total Assessment (%)	Description
Written Examination	60	3-hour written examination
Continuous Assessment	30	Assignments
Practical Work	10	Laboratory work

Course and Assessment Policy

Late Submission / Missed quiz
As per Departmental Regulations.

Academic Conduct
The University Regulations on academic misconduct will be strictly enforced. Students are strongly recommended to refer to http://www.hku.hk/student/plagiarism/ for further details. In addition, students should familiarise themselves with http://lib.hku.hk/turnitin/.

Means / Processes for Student Feedback on Course

SFTL
Students are asked to complete this evaluation (Student Feedback on Evaluation of Teaching and Learning) of their learning experiences at the conclusion of the course. Questionnaire items relate to the overall evaluation of the course as well as an evaluation of teaching.

Other means of student feedback
In addition, individual teachers may seek student feedback during class discussions and consultation after class.

Date: Date: July 2026 Completed by: Prof. J.J. Wang

Learning Resources

Required text / References

1. Gere, J.M. and Goodno, B.J., Mechanics of Materials, 8th Edition, Cengage Learning, 2013. (Recommended)
2. Beer, F.P., Johnston, E.R. and DeWolf, J.T., Mechanics of Materials, 6th Edition, McGraw Hill, 2013.
3. Hibbeler, R.C., Engineering Mechanics: Static & Dynamics, 13th Edition, Pearson, 2013
4. Hannah, J. and Hillier, M.J., Applied Mechanics, Pitman, 1995
5. Riley, W.F., Sturges, L.D. and Morris, D.H., Statics and Mechanics of Materials, Wiley, 1995.

Course website

Selected course materials will be posted on Moodle in addition to announcements of important dates/deadlines: students are advised to regularly consult the Moodle of this course.